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Bachelor Thesis

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Investigating the Optimal Geometry of a Vanadium Calibration Sample for Neutron Scattering

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Abstract

Contents

Abstract **ii**

1 Introduction **1**
1.1 Motivation and Background . . . 1
1.2 Neutron scattering and vanadium calibration . . . 1
1.3 The Union concept in McStas . . . 1
1.4 Project objectives . . . 1

2 Theoretical background **2**
2.1 Neutron transport and Monte Carlo method . . . 2
2.2 Scattering and absorption cross sections . . . 2
2.3 Multiple scattering theory . . . 2
2.4 Angular distributions and 4π monitors . . . 2
2.5 Intensity normalization and solid angle effects . . . 2

3 Development of MCmini - A minimal C-based Monte Carlo neutron simulation program **3**
3.1 Design philosophy and purpose . . . 3
3.2 Implementation of a simple source . . . 3
3.2.1 Wavelength distribution comparison . . . 4
3.2.2 Spatial distribution comparison . . . 5
3.2.3 Normalization and neutron weight . . . 5
3.3 Monitor implementations . . . 7
3.3.1 4π spherical monitor . . . 7
3.4 Geometry implementations . . . 8
3.4.1 Sphere . . . 8
3.4.2 Cylinder . . . 8
3.5 Multiple scattering algorithm . . . 8

4 Validation of Union in McStas **9**
4.1 Validation Strategy . . . 10
4.1.1 What is matched, Metrics, and Statistical treatment . . . 10
4.2 `Source_simple` comparison and Validation . . . 10
4.3 Solid Sphere: Multiple Scattering Validation . . . 10
4.3.1 Spatial intensity comparison . . . 10
4.3.2 Integrated intensity . . . 10
4.3.3 Cosine correction . . . 10
4.3.4 Polar suppression analysis . . . 11

4.4 Solid Cylinder Validation 11
4.4.1 Small Cylinder, (R = to be determined, H = TBD) 11
4.4.2 Medium Cylinder, (R = 5 mm, H = 2 cm) 11
4.4.3 Large Cylinder, (R = TBD, H = TBD) 11
4.4.4 Geometric interpretation 11
4.5 Scatter-Order Decomposition 11
4.5.1 1–5 scatter comparison 11
4.5.2 Quantitative agreement 12
4.5.3 High-order statistics 12
4.6 Summary of Validation Results 12
4.6.1 Agreement levels 12
4.6.2 Remaining discrepancies 12
4.6.3 Confidence in Union implementation 12

5 Experimental Studies and Geometry Optimization in McStas **13**
5.1 Objective of the Optimization Study 13
5.2 Simulation Setup in McStas 13
5.3 Performance Metrics 13
5.4 Thickness Study 13
5.5 Radius and Height Study 13
5.6 Optimization Strategy 13
5.7 Recommended Geometry for Calibration Use 13

6 Discussion **14**
6.1 Physical interpretation of trends 14
6.2 Limitations of Monte Carlo model 14
6.3 Statistical precision 14
6.4 Limits of Union 14

7 Conclusion **15**
7.1 Validation summary 15
7.2 Optimization results 15
7.3 Future work 15

Chapter 1

Introduction

1.1 Motivation and Background

1.2 Neutron scattering and vanadium calibration

1.3 The Union concept in McStas

1.4 Project objectives

Chapter 2

Theoretical background

2.1 Neutron transport and Monte Carlo method

2.2 Scattering and absorption cross sections

2.3 Multiple scattering theory

2.4 Angular distributions and 4π monitors

2.5 Intensity normalization and solid angle effects

Chapter 3

Development of MCmini - A minimal C-based Monte Carlo neutron simulation program

3.1 Design philosophy and purpose

MCmini is an independently developed minimal Monte Carlo neutron transport system implemented in C. MCmini is simple by design, with the core philosophy being to reduce the primary physical interactions down to its core elements to best recreate the results of a real experiment. Its purpose is to validate and reference the result of the multiple scattering experiment performed with Union in McStas.

3.2 Implementation of a simple source

The source that will be implemented in MCmini will follow the same input parameters as `Source_simple` in McStas[1]. In order to validate a simple source in MCmini a direct comparison to `Source_simple` in McStas was made. The goal was to ensure that spatial and wavelength distribution of the source in MCmini matched that of McStas.

Source definition

Each neutron is generated in two steps:

1. **Source position sampling** The emission point (x, y, z) is sampled uniformly either:
   - inside a circular disk of radius R, or
   - inside a rectangular area of width x_{width} and height y_{height}.
2. **Target sampling and direction** A random target point is sampled uniformly inside a rectangular window of size

$$\text{focus\_xw} \times \text{focus\_yh}$$

located at distance `dist` along the z-axis.

The neutron direction is then defined as the normalized vector

$$\vec{v} = \frac{\vec{r}_{\text{target}} - \vec{r}_{\text{source}}}{|\vec{r}_{\text{target}} - \vec{r}_{\text{source}}|}.$$

with this we define rays with:

$$r(t) = \vec{r_0} + \vec{v} \cdot t \tag{3.1}$$

The following parameters were used in both McStas and `mcmini`:

- Number of neutrons: $n = 1 \cdot 10^6$
- Source radius: $r = 0.1$ m
- Target width: $\text{focus\_xw} = 0.4$ m
- Target height: $\text{focus\_yh} = 0.4$ m
- Distance to target: $\text{dist} = 1.0$ m
- Wavelength center: $\lambda_0 = 4.0$ Å
- Wavelength half-spread: $\Delta\lambda = 0.1$ Å
- Divergence mode: uniform ($\texttt{gauss} = 0$)

Wavelength sampling

The particles are sampled uniformly with $\Delta\lambda$ around λ_0, that is:

$$\lambda \sim \mathcal{U}\left[\lambda_0 - \Delta\lambda,\ \lambda_0 + \Delta\lambda\right]. \tag{3.2}$$

For $\lambda_0 = 4.0$ Å and $\Delta\lambda = 0.1$ Å, this yields

$$\lambda \in [3.9,\ 4.1] \text{ Å}, \tag{3.3}$$

3.2.1 Wavelength distribution comparison

Monitor definition

The monitor that will check the spatial distribution is a flat monitor. The monitor is placed at $(0, 0, 1)$ with $\texttt{x\_width} = 0.5$ m and $\texttt{y\_height} = 0.5$ m. To find intersection solve for time when z-coordinate is equal to 1 and check for that time if the rays is within the monitor plane. How the monitor works will be explained in greater detail later. idea behind the simple monitor is simple enough to suffice with this explanation.

3.2.2 Spatial distribution comparison

First is the spatial comparison. MCmini bins the number of ray intersection within `x_width` $= 0.5$ m and `y_height` $= 0.5$ m into $(200, 200)$ array. Figure 3.1 shows the comparison between MCmini raw counts and McStas with same parameters. We will not force same color scare for now.

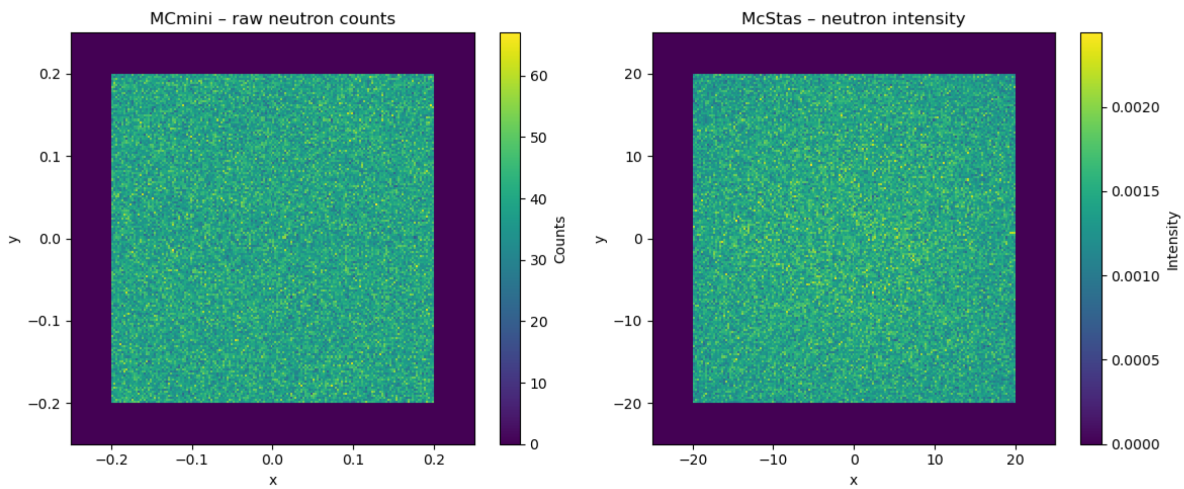


Figure 3.1: McStas `Source_div` intensity map on flat monitor.

3.2.3 Normalization and neutron weight

In MCmini the neutron weight is factorized as

$$p = p_{\text{mul}} \cdot p_{\text{dir}}, \tag{3.4}$$

in direct analogy with the `Source_simple` component in McStas.

Global source factor p_{mul}

The factor p_{mul} sets the overall normalization of the source and depends only on global source parameters:

$$p_{\text{mul}} = \begin{cases} \Phi \cdot 10^4 \cdot A_{\text{src}} \cdot 2\Delta\lambda, & \text{if wavelength spread is used} \\ \Phi \cdot 10^4 \cdot A_{\text{src}} \cdot 2\Delta E, & \text{if energy spread is used} \\ \dfrac{1}{4\pi}, & \text{if } \Phi = 0 \end{cases} \tag{3.5}$$

where

$$A_{\text{src}} = \begin{cases} \pi R^2, & \text{circular source} \\ x_{\text{width}}\, y_{\text{height}}, & \text{rectangular source.} \end{cases} \tag{3.6}$$

This factor is identical for all neutrons in a run with fixed source parameters. Note that in McStas p_{mul} contains a division by N_{count}, whereas in MCmini this division is performed later at monitor level (per-history normalization). Therefore the $1/N_{\text{count}}$ factor is intentionally omitted here.

Directional factor p_{dir}

The directional weight accounts for the probability that a sampled ray intersects a finite target area. For a target centered along the $+z$ axis,

$$p_{\text{dir}} = \frac{A_{\text{focus}}}{d^2} \cos^2 \theta, \tag{3.7}$$

where

$$A_{\text{focus}} = x_{\text{focus}}\, y_{\text{focus}}, \tag{3.8}$$

$$d = \text{distance from sampled source point to sampled target point}, \tag{3.9}$$

$$\cos\theta = \frac{\Delta z}{d}. \tag{3.10}$$

The $\cos^2\theta$ term corresponds to `order=2` in `Source_simple`. Unlike p_{mul}, p_{dir} varies for each neutron history because both d and θ depend on the sampled source and target coordinates.

Monitor intensity

so that the reported *intensity* per pixel is simply

$$I_{ij} = \sum_{\text{hits}} p_{\text{mul}}\, p_{\text{dir}} \cdot \frac{1}{N_{count}}. \tag{3.11}$$

No geometrical solid-angle normalization is applied internally. Thus the raw monitor signal corresponds to the weighted sum of neutron probabilities per pixel.

Figure 3.2

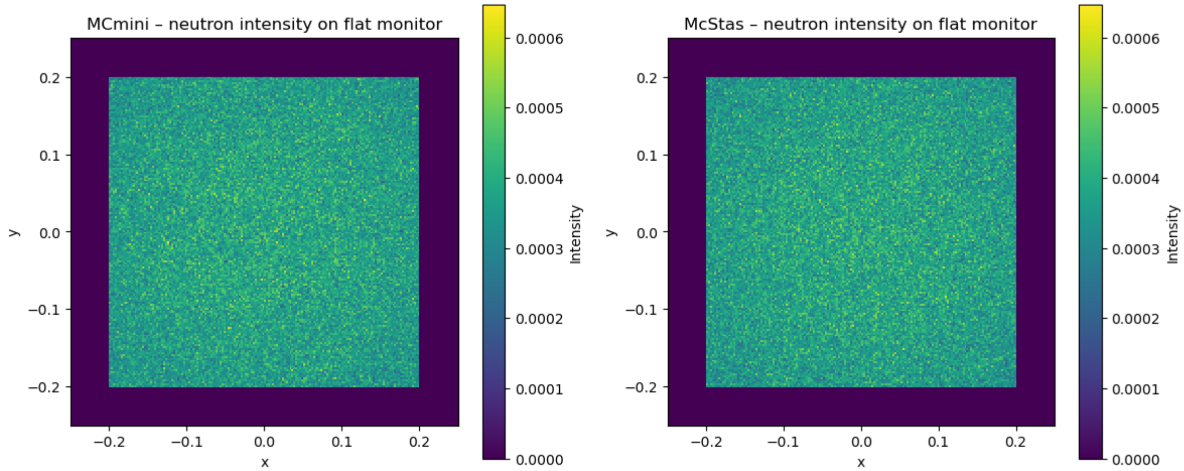


Figure 3.2: McStas `Source_simple` intensity map on flat monitor.

Figure 3.3

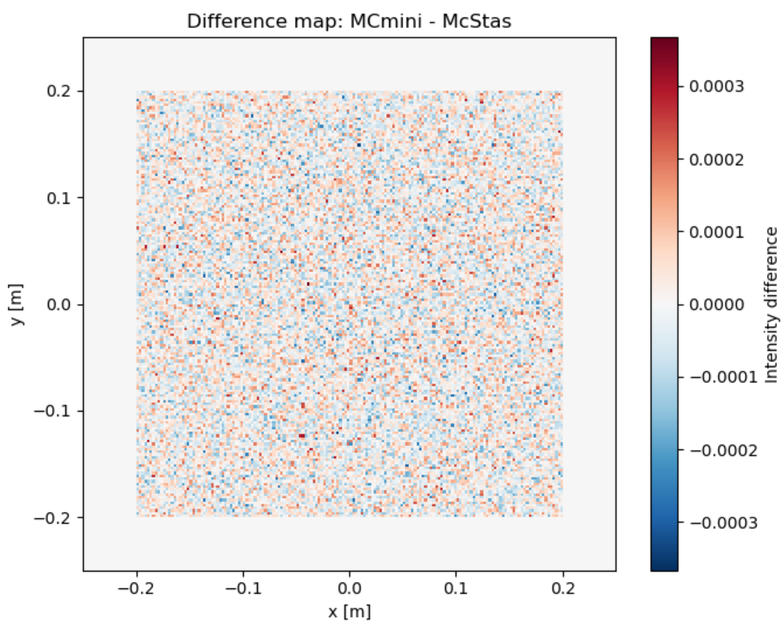


Figure 3.3: McStas `Source_simple` intensity map on flat monitor.

3.3 Monitor implementations

3.3.1 4π spherical monitor

The same logic used in the flat monitor is used in the 4PI monitor. The equations to be solve are different and we will go more in depth here with respect to how data is binned and such

3.4 Geometry implementations

3.4.1 Sphere

Sphere is simplest to solve so that will function as our first example.

3.4.2 Cylinder

Cylinder is slightly more complicated, as we have to sort intersection trough lid and wall. A solid cylinder will suffice in the comparison to McStas

3.5 Multiple scattering algorithm

As long as particle is alive the ray intersection functions are called after a scattering event. Absorbed rays are killed and unlike McStas which weighs the probability that the ray continues to reduce variance, MCmini simply removes the ray completely.

Chapter 4

Validation of Union in McStas

4.1 Validation Strategy

4.1.1 What is matched, Metrics, and Statistical treatment

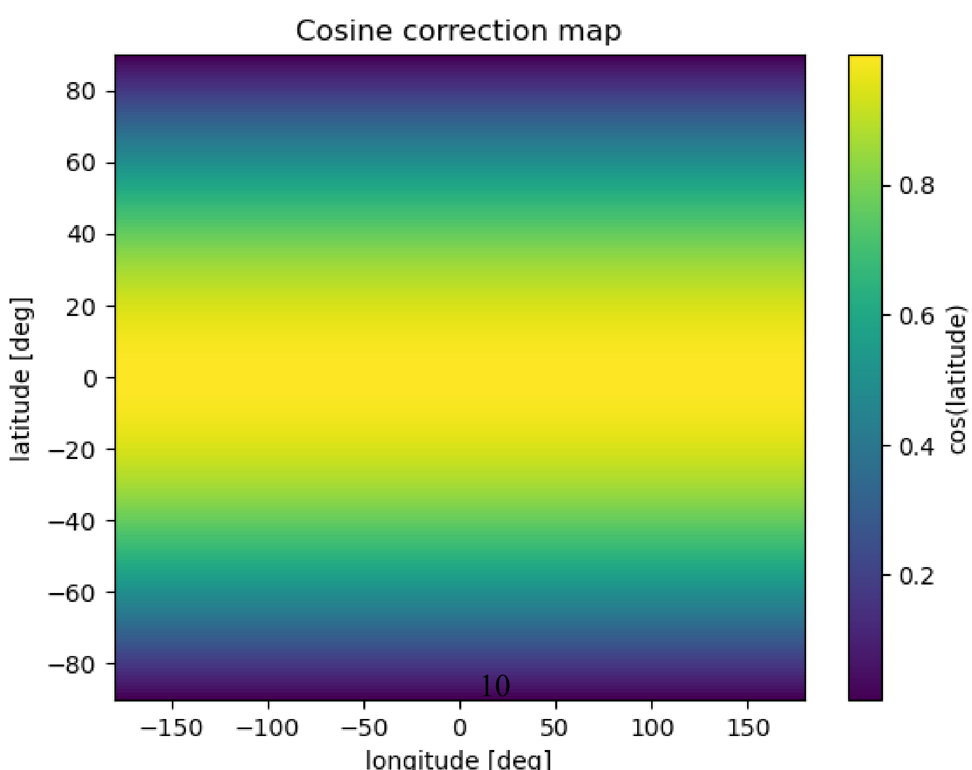
4.2 `Source_simple` comparison and Validation

4.3 Solid Sphere: Multiple Scattering Validation

4.3.1 Spatial intensity comparison

4.3.2 Integrated intensity

4.3.3 Cosine correction



4.3.4 Polar suppression analysis

4.4 Solid Cylinder Validation

4.4.1 Small Cylinder, (R = to be determined, H = TBD)

4.4.2 Medium Cylinder, (R = 5 mm, H = 2 cm)

4.4.3 Large Cylinder, (R = TBD, H = TBD)

4.4.4 Geometric interpretation

4.5 Scatter-Order Decomposition

4.5.1 1–5 scatter comparison

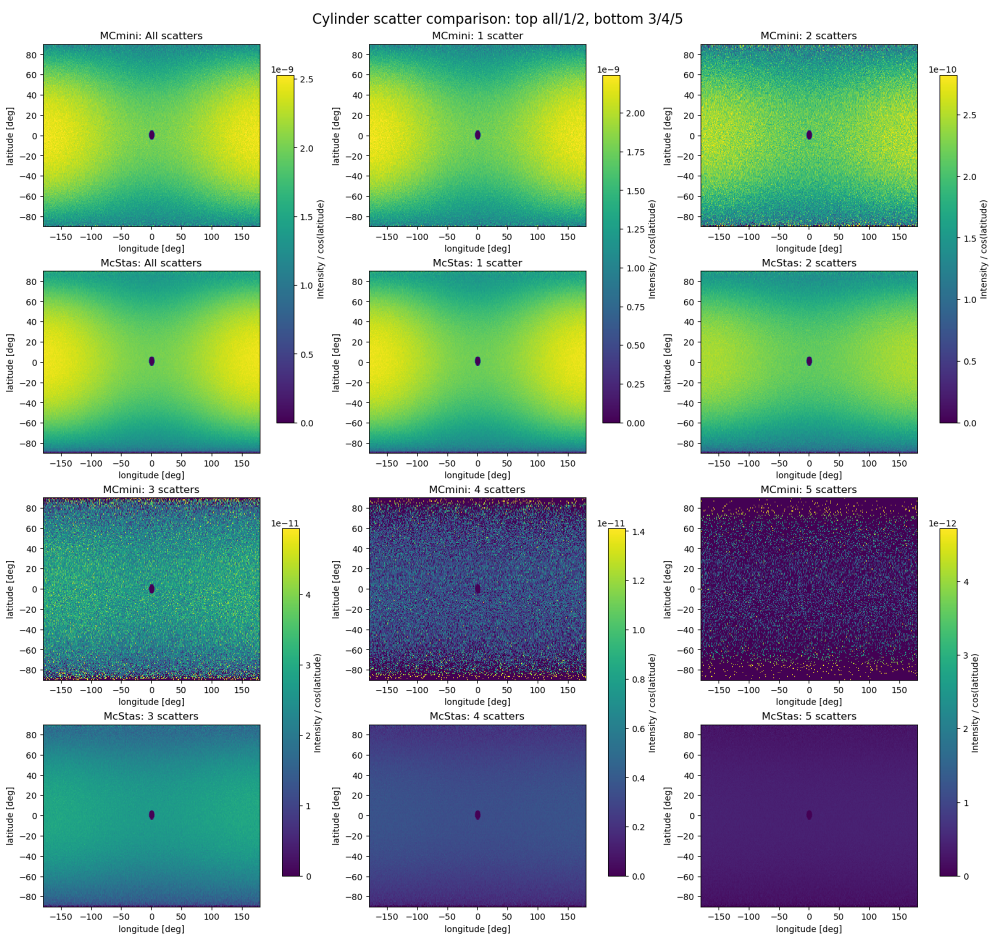


Figure 4.2: Enter Caption

4.5.2 Quantitative agreement

4.5.3 High-order statistics

4.6 Summary of Validation Results

4.6.1 Agreement levels

4.6.2 Remaining discrepancies

4.6.3 Confidence in Union implementation

Chapter 5

Experimental Studies and Geometry Optimization in McStas

5.1 Objective of the Optimization Study

5.2 Simulation Setup in McStas

5.3 Performance Metrics

5.4 Thickness Study

5.5 Radius and Height Study

5.6 Optimization Strategy

5.7 Recommended Geometry for Calibration Use

Chapter 6

Discussion

6.1 Physical interpretation of trends

6.2 Limitations of Monte Carlo model

6.3 Statistical precision

6.4 Limits of Union

Chapter 7

Conclusion

7.1 Validation summary

7.2 Optimization results

7.3 Future work

Bibliography

[1] McStas Development Team. *Source_simple Component Documentation, McStas 3.6.5*. https://www.mcstas.org/download/components/3.6.5/sources/Source_simple.html. Accessed: 3 March 2026. 2024.